Business field: AT SE RS

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Cross Reference

The following table shows a cross reference between the Siemens and the customer type identifications for the SDM Airbag Versions SABx Opel :

Siemens MLFB	Customer Part Number	Alpha Code / Snap ID	Customer Target Platform	SOP	Remarks
5WK4 2925	09 229 302	BF / 0Dh	T-Car 3050 Astra	MY '00.75	SAB8X 1)
5WK4 2926	09 229 303	BG / 0Dh	T-Van 0601 Zafira	MY '00.75	SAB8X 1)
5WK4 2927	09 229 304	BH / 0Dh	T-Coupe 3016 Coupe	MY '00.75	SAB8X 1)

1) : Deployment loops #1 to #8
Output for "Deployment Commanded" (DCO)
AOS1 interface and AOS lamp driver *)
SABx specific connector pin-out (28 pins)
Sensor improvement : „Omega“ Wingback sensor

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1 Introduction

1.1 Basic documents

The following GME documents have been used as the basis for this specification :

- OPEL QT 12 6543 part1 05/93 issue 07. September 1993 Section. 7.6.1
  Eigendiagnose für elektronische Systeme mit Mikroprozessoren
- GME 12 543 part 4 09/94 issue 30.09.1994
  Self Diagnosis of Microprocessor Controlled Electronic Systems
  Communication Structure to Keyword 82
- GME 12 543 part 5 05/95 issue 1
  Self Diagnosis for Electronic Systems with Microprocessors
  Part 5 : Physical Layer and Data Link Layer (200 Baud Initialization)

The interface has been implemented as a bi-directional single wire interface.

1.2 Definitions

Block: A block is the smallest repeatable unit for data transfer and always has the same structure.

Telegram: The sum of all blocks that are exchanged during the data transfer.

Master: The party in the communication that sends a block.

Slave: The receiver of a block.

In the following text, often repeated terms are abbreviated :

- Communication participants :
  Tester : T
  Electronic Control Unit : ECU
- Numerical formats (adjacent to numbers or variables) :
  Binary Number : B
  Decimal Number : D
  Hexadecimal Number : H

In addition, all time values specified in the text are shown symbolically (e.g T_{Idle}, w0). The definition of all these values can be found in a quick overview in the rear part of this document.

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2 Communication Sequence

2.1 Communication Stimulus

A stimulus of the serial data communication is only possible after an "Idle Line" ("High" level on the data line) of at least T_{idle}.

The stimulus is sent by the tester which transfers an address word in 10-bit format with one of the following baud rates :

Baud rate = 200 bd or 5 bd (± 2 %)

Word format of the address :

1	Start bit	logical level "0"
8	Data bits	LSB first
1	Stop bit	logical level "1"

Address word for the Airbag ECU : 62H

The Airbag ECU aborts the stimulus detection when :

- no "Idle Line" is detected
- no valid start bit is detected
- the received address does not correspond to the Airbag ECU address
- no valid stop bit is detected

The Airbag ECU is capable to automatically detect the address word baud rate and detect high speed communication also during "Idle Line" , start bit , address and stop bit detection.

2.2 Communication Initiation

Initiation of the communication is carried out according to the following scheme:

- After a wait time w0 after receiving the stimulus (= end of the stop bit), the Airbag ECU sends the synchronization byte 55H which informs the tester of the baud rate (9600 bd ± 200 bd) and the voltage level.

  Word format of the synchronization byte :

1	Start bit	logical level "0"
8	Data bits	LSB first
1	Stop bit	logical level "1"

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- After a further wait time w1, the Airbag ECU sends two key bytes which specify the communication parameters for the serial data transfer to follow.

  Word format of the communication :

1	Start bit	logical level "0"
8	Data bits	LSB first
1	Stop bit	logical level "1"

  Word format of the key bytes:

1	Start bit	logical level "0"
7	Data bits	LSB first
1	Parity bit	(Parity odd)
1	Stop bit	logical level "1"

  Content of the key bytes:

1st	Key byte = 52H
2nd	Key byte = 80H

- The wait time between the 1st and 2nd key byte is w2.

- If the one's complement of the 2nd key byte is received correctly, the ECU enters the operating mode ‘Diagnosis communication’ and after t1 sends the identification block. Whilst in this operating mode, the ECU is no longer capable of deploying any restraint system. This state is indicated by a constant activation of the warning lamp.

2.3 Communication Structure of Keyword 82

The data transfer occurs using a block structure. A block is the smallest repeatable unit for transfer and always has the same structure. The transfer of a block occurs without any handshaking procedure. Transfer safeguarding is provided by means of two checksum bytes.

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3 Telegram Structure

A Telegram is defined as the sum of all transfered blocks during a communication. It begins with the output of the identification block by the ECU (after receiving the complement of the 2nd key byte) and has the following structure:

Transmitter	Receiver	Block
ECU	T	Transmit ECU identification block
	.	
	.	
	.	
T	ECU	Request block (e.g.read snapshot)
ECU	T	Response block (e.g. transmit snapshot)
	.	
	.	
	.	
T	ECU	End of diagnosis
ECU	T	One status block

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4 Block Structure

Byte	Description	Content
1	Block length	n
2	Block title	
3	Data byte 1 (if required)	
.		
.		
.		
n	Data byte n-2 (if req'd)	
n+1	Checksum (high byte)	
n+2	Checksum (low byte)	

Block length: Number of bytes in the block.
The two checksum bytes are not included in the count.
The maximum block length 'n' ist 42 bytes.

Block title: Encodes the block type.
The block types used are listed in Section 6 .

Checksum: The 16 Bit Checksum is generated by the addition of bytes 1 .. n .
The checksum bytes are not included in the calculation.
On receiving the block, the ECU calculates the checksum and compares the value with the transfered checksum bytes.
Only blocks with correct checksums are processed.

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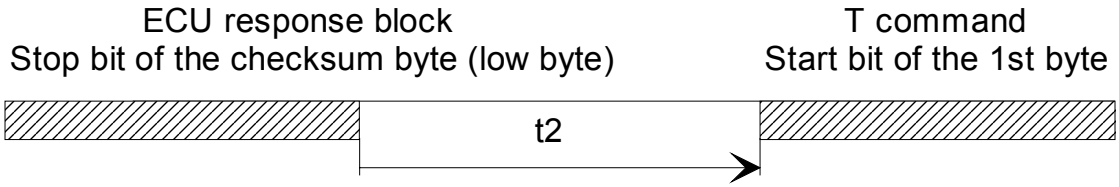
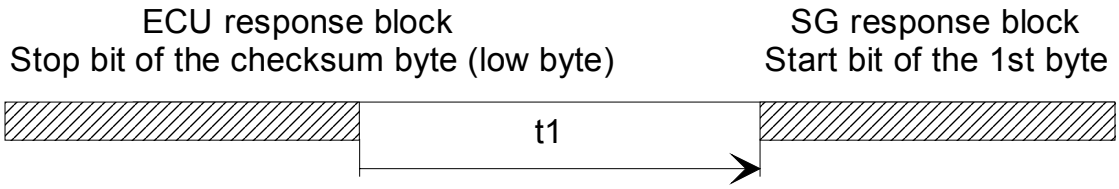
5 Data Transfer

After communication initiation, the ECU commences with the transmission of the block "Transmit ECU identification". Each response block is repeated until the tester transfers a new request block. For the change of the master function, the following time conditions apply:



t2 : Intervention window for the Tester



t1 : Timeout for repetition of response block

If the tester does not commence with the transfer of a block during the time window t2, then the last response block from the Control Unit is repeated (with updated contents) after elapse of the time window t1. The following conditions apply for the value of t1 :

Before repeating the response blocks "Transmit snapshot" and "Transmit trouble codes", the ECU performs a self test cycle of approx. 400 ms (max. 500 ms) duration. For this reason, one or more "Wait Tester" status blocks requesting an increase in the block-timeout are sent before the new output of the response block. The time window t1 which immediately follows the "Wait Tester" status block is thereby extended to max. t1' . A similar wait time is requested in conjunction with the commands "Clear trouble codes" and "Control ECU" .

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6 Data Blocks

This section describes the blocks which are available for the communication between tester and the Airbag ECU .

6.1 Overview

The ECU responds to a tester request or command block with the associated response block :

Request blocks

Tester		ECU	Block
Block title	**Function**	**Block title**	**Function**
10H	Read ECU Identification	A0H	Transmit ECU Identification
11H	Read snapshot	A1H F0H	Transmit snapshot Status block "Wait Tester"
12H	Read trouble codes	A2H F0H	Transmit trouble codes Status block "Wait Tester"
21H	Read EEPROM cell contents	ABH F0H	Transmit EEPROM cell contents Status block "Wait Tester"

Command blocks

Tester		ECU	Block
Block title	**Function**	**Block title**	**Function**
B0H	Read status block	F0H	Status block "Acknowledge"
B1H	Clear trouble codes	F0H F0H	Status block "Wait Tester" Status block "(No) Acknowledge"
B2H	End of diagnosis	F0H	Status block "Acknowledge"
BCH	Control ECU	F0H	Status block "(No) Acknowledge"

6.2 Request Blocks

In the following description, each tester block is presented with the corresponding response block. The response is repeated cyclically.

The response of the ECU in case of transfer errors is described in chapter 9 .

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6.2.1 ECU Identification

Tester block :

Byte	Description	Content
1	Block length	02H
2	Block title:	
	Read ECU Identification	10H
3	Checksum (high byte)	00H
4	Checksum (low byte)	12H

ECU response :

The content of the ECU response is device type specific. The ECU identification contains:

- the Alpha-Code,
- the OPEL Part No.,
- the SIEMENS HW - No.,
- the SIEMENS SW - No.,
- the Production Date (Week / Year)

The SIEMENS HW-No. matches the barcode on the ECU (except for the 2-character alpha code and the checksum character) and thereby together with the alpha code allows each specific device to be individually identified.

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ECU Identification (generic contents) :

Byte	Description		Content	Remark	
1	Block length		22H		
2	Block title : Transmit ECU Identification		A0H		
3	1st ASCII character	"t"	ttH	Alpha-Code	(2 bytes)
4	2nd ASCII character	"t"	ttH	type specific	
5	3rd ASCII character	" "	20H	Delimiter	
6	4th ASCII character	"t"	ttH	OPEL Part No.	(8 bytes)
7	5th ASCII character	"t"	ttH	type specific	
8	6th ASCII character	"t"	ttH		
9	7th ASCII character	"t"	ttH		
10	8th ASCII character	"t"	ttH		
11	9th ASCII character	"t"	ttH		
12	10th ASCII character	"t"	ttH		
13	11th ASCII character	"t"	ttH		
14	12th ASCII character	" "	20H	Delimiter	
17	13th ASCII character	"S"	53H	Siemens HW-No.	(8 bytes)
18	14th ASCII character	"n"	nnH	Snnnnnnn	
	...				
22	20th ASCII character	"n"	nnH		
23	21st ASCII character	" "	20H	Delimiter	
24	22nd ASCII character	"x"	xxH	Siemens SW-No.	(6 bytes)
	...			xxxxxx	
29	27th ASCII character	"x"	xxH		
30	28th ASCII character	" "	20H	Delimiter	
31	29th ASCII character	"w"	wwH	Production date	(4 bytes)
32	30th ASCII character	"w"	wwH	wwyy	
33	31st ASCII character	"y"	yyH		
34	32nd ASCII character	"y"	yyH		
35	Checksum (high byte)		ppH		
36	Checksum (low byte)		ppH		

"t" ttH : type specific identification entries (see following pages)

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ECU Identification for T-0501 Car MY '01
(SAB8X , MLFB 5WK4 2925 , OPEL Part No. 09 229 302) :

Byte	Description		Content	Remark	
3	1st ASCII character	"B"	42H	Alpha-Code	(2 bytes)
4	2nd ASCII character	"F"	46H	BF	
6	4th ASCII character	"0"	30H	OPEL Part No.	(8 bytes)
7	5th ASCII character	"9"	39H	09 229 302	
8	6th ASCII character	"2"	32H		
9	7th ASCII character	"2"	32H		
10	8th ASCII character	"9"	39H		
11	9th ASCII character	"3"	33H		
12	10th ASCII character	"0"	30H		
13	11th ASCII character	"2"	32H		

ECU Identification for T-0601 Car MY '01
(SAB8X , MLFB 5WK4 2926 , OPEL Part No. 09 229 303) :

Byte	Description		Content	Remark	
3	1st ASCII character	"B"	42H	Alpha-Code	(2 bytes)
4	2nd ASCII character	"G"	47H	BG	
6	4th ASCII character	"0"	30H	OPEL Part No.	(8 bytes)
7	5th ASCII character	"9"	39H	09 229 303	
8	6th ASCII character	"2"	32H		
9	7th ASCII character	"2"	32H		
10	8th ASCII character	"9"	39H		
11	9th ASCII character	"3"	33H		
12	10th ASCII character	"0"	30H		
13	11th ASCII character	"3"	33H		

ECU Identification for T-3016 Car MY '01
(SAB8X , MLFB 5WK4 2927 , OPEL Part No. 09 229 304) :

Byte	Description		Content	Remark	
3	1st ASCII character	"B"	42H	Alpha-Code	(2 bytes)
4	2nd ASCII character	"H"	48H	BH	
6	4th ASCII character	"0"	30H	OPEL Part No.	(8 bytes)
7	5th ASCII character	"9"	39H	09 229 303	
8	6th ASCII character	"2"	32H		
9	7th ASCII character	"2"	32H		
10	8th ASCII character	"9"	39H		
11	9th ASCII character	"3"	33H		
12	10th ASCII character	"0"	30H		
13	11th ASCII character	"4"	34H		

6.2.2 Snapshot

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6.2.2.1 Message structure

Tester block :

Byte	Description	Content
1	Block length	02H
2	Block title:	
	Read Snapshot	11H
3	Checksum (high byte)	00H
4	Checksum (low byte)	13H

ECU response:

The ECU response contains :

- all 16 entries in the trouble code memory together with their respective status,
- the System Configuration Byte
- the Resistance Values for all deployment loops (number hardware dependent)
- the Error Timer
- the Activation Status Byte
- the Peripheral Status Bytes (number hardware dependent)
- the Supply Voltage Value

The ECU registers up to 16 trouble codes, all 16 entries will be contained in the snapshot. A complete readout of trouble codes can also be obtained with the "Read trouble codes" request.

Before repeating the response blocks "Transmit snapshot", the ECU performs a self test cycle of approx. 400 ms (max. 500 ms) duration. For this reason, "Wait Tester" status blocks are used to keep inter-block-times to their specified limits :

Several "Wait tester" status blocks requesting block-timeouts of 250 ms are sent before the new output of the response block.

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Snapshot Contents :

Byte	Description		Content
1	Block length		2AH
2	Block title		
	Transmit snapshot		A1H
3	**Snapshot ID**		**10H**
4	1st Trouble code byte		yyD
5	2nd Trouble code byte		yyD
6	3rd Trouble code byte		yyD
7	4th Trouble code byte		yyD
8	1st Trouble code status byte	Bits	B4 A4 B3 A3 B2 A2 B1 A1
9	5th Trouble code byte		yyD
10	6th Trouble code byte		yyD
11	7th Trouble code byte		yyD
12	8th Trouble code byte		yyD
13	2nd Trouble code status byte	Bits	B8 A8 B7 A7 B6 A6 B5 A5
14	9th Trouble code byte		yyD
15	10th Trouble code byte		yyD
16	11th Trouble code byte		yyD
17	12th Trouble code byte		yyD
18	3rd Trouble code status byte	Bits	B12 A12 B11 A11 B10 A10 B9 A9
19	13th Trouble code byte		yyD
20	14th Trouble code byte		yyD
21	15th. Trouble code byte		yyD
22	16th Trouble code byte		yyD
23	4th Trouble code status byte	Bits	B16 A16 B15 A15 B14 A14 B13 A13
24	System Configuration Byte		ccH
25	Resistance value FAD		rrH
26	Resistance value FAP		rrH
27	Resistance value BPD		rrH
28	Resistance value BPP		rrH
29	Resistance value SAD		rrH
30	Resistance value SAP		rrH
31	Resistance value CUD		rrH
32	Resistance value CUP		rrH
33	Error timer (MSB)		ttH
34	Error timer (LSB)		ttH
35	Activation Status Byte		aaaaaaaaB
36	1st Peripheral Status Byte		ssssssssB
37	2nd Peripheral Status Byte		ssssssssB
38	3rd Peripheral Status Byte		ssssssssB
39	4th Peripheral Status Byte		ssssssssB
40	5th Peripheral Status Byte		ssssssssB
41	6th Peripheral Status Byte		ssssssssB
42	Supply voltage value		vvH
43	Checksum (high byte)		ppH
44	Checksum (low byte)		ppH

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6.2.2.2 Message contents

Trouble codes :

The trouble codes generated by the Airbag ECU are transmitted in decimal form (value range 0..255) .

The trouble codes are described in chapter 7 .

Trouble code status bytes :

Each trouble code byte is assigned to 2 bits in a status byte to indicate the error status :

Bit: Bx	Ax	Meaning:	Warning lamp:
0	0	trouble code byte unoccupied	off
0	1	Error is stored in EEPROM and currently not present	on *)
1	0	not possible	
1	1	Error is stored in EEPROM and currently present	on **)

*) : for trouble codes
- with "NWL" status or
- with "LAO" status and error never present in the current operating cycle
- with "LPO" status and error not currently present

(see chapter 7) : off

**): for trouble codes with "NWL" status (see chapter 7) : off

Note : The warning lamp status will be determined from all entries in the active trouble code memory block and will also be activated for “Supply voltage too low” or “Supply voltage too high” conditions .

System Configuration Byte :

The "System Configuration Byte" reflects the last parameter value used for programming the SRS configuration (see chapter "Control ECU").

As long as no programming has been performed, the parameter value will show as 00H .

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Resistance values (deployment loops) :

The measured values are transmitted as unsigned hexadecimal integers. The resistance value for not configured deployment circuits will always be reported as 00H . The resistance measurement is only active after configuration programming (trouble code 52 no longer present) and as long as there are no shorts to battery or ground present in any deployment loop.

The hexadecimal integer measurement value may be converted to a display value by means of the following formula:

Display value = (Transmitted value / 34) * Ohm

The real value may deviate from the display value within the following range :

Real value = (Display value ± 15 %) ± 0.35 Ohm

Instead of display values of 6.6 Ohm and beyond (transmitted values E0H to FFH) the display should read "open loop".

The threshold values for the resistance related trouble codes are set to :
(configured loops only, typical resistance thresholds given in brackets)

	Resistance too low	Resistance too high
FAD	40H (1.9 Ohm)	A7H (4.9 Ohm)
FAP, BPD and BPP, SAD and SAP, CUD and CUP	2CH (1.3 Ohm)	96H (4.4 Ohm)

N.B. : These four threshold values are programmable in non-volatile memory at the supplier's assembly line and may be subject to change if required by the customer.

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Error timer :

The error timer shows the duration of warning lamp activation as a result of registered trouble codes.

The error timer value is stored in EEPROM and kept after ignition off or power loss.

The error timer is incremented if the warning lamp is activated caused by trouble codes stored in EEPROM. There is no roll over.

The actual error timer value is stored within every deployment record.

The command "Clear Trouble Codes" resets the error timer provided no trouble code with "NC" status has been stored.

The LSB is incremented from a start value of 00H every 10 s to a value of F0H (= 40 min). When the LSB maximum value is achieved, the MSB is incremented every 10 min to a value of F0H (= 40 hours), while the LSB will stay at its maximum value of F0H.

Activation Status Byte :

This byte displays the number of trouble codes 67 to 69 present in trouble code memory blocks flagged as reset due to a "Clear Trouble Codes" command. Each of these trouble codes can be reset a maximum of two times.

Furthermore, this byte contains a flag showing whether any of the trouble codes 65 to 69 is stored in the active trouble code memory block.

The "Activation Status Byte" is constructed as follows:

Bit 0 :	Trouble Code 67	1 = reset once after activation
Bit 1 :	Trouble Code 67	1 = reset twice after activation
Bit 2 :	Trouble Code 68	1 = reset once after activation
Bit 3 :	Trouble Code 68	1 = reset twice after activation
Bit 4 :	Trouble Code 69	1 = reset once after activation
Bit 5 :	Trouble Code 69	1 = reset twice after activation
Bit 6 :	unused	always = 0
Bit 7 :	Deployment flag	0 = no deployment 1 = Trouble Code 65 to 69 present

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Peripheral Status Bytes :

The peripheral status bytes are only valid after configuration programming (trouble code 52 no longer present) .

1st Peripheral Status Byte : Lamp & Supply Voltage Status Byte

Bit 0 .. 4 :	Counter for AOS1 Tdly1	typ. resolution 400 ms, max. value 10
	The counter starts at 0 and counts up to / stops at the max. value (Tdly1 expired).	
N.B. :	The max. value is programmable in non-volatile memory at the supplier's assembly line and may be subject to change in the range of 0 to 30 if required by the customer.	
Bit 5 :	AOS status lamp	0 = off 1 = on
Bit 6 :	Over- / Undervoltage condition	0 = normal supply voltage present 1 = over- / undervoltage present
Bit 7 :	SRS warning lamp	1 = on (always in diagnostic mode)

2nd Peripheral Status Byte : AOS1 Status Byte

- With low-speed AOS communication (AOS1) present:

  Bit 0 .. 6 : Bit 6 .. 0 of last correct PPD/CPOD/AOS status message
  N.B. : The bit order is reverse to the message order.
  The contents of this data field will reflect the last message received, which during Tdly1 not necessarily reflects the message that has led to the actual setting of the deployment readiness in Bit 7 .

  Bit 7 : Deployment readiness of passenger side frontal and side airbag 0 = yes
  1 = no

  AOS1 configured:
  The value is initialised to FFH at power ON and after 4 expirations of Tdly2 .
  Before this condition is met the contents reflect the last message received.

  AOS1 not configured:
  The value is initially with 00H. As soon as the 1st AOS message had been received the value is initialised with FFh and afterwards reinitialised to FFH after 4 expirations of Tdly2 (see also 5th peripheral status byte).

- With AOS1 not present and not configured:

  The value remains set to 00H .

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3rd Peripheral Status Byte : Side Impact Sensor Status Driver Side and
4th Peripheral Status Byte : Side Impact Sensor Status Passenger Side

A status byte is provided for each sensor satellite of the side airbag system.
Status information is provided by means of status bits (active status = 1) as follows :

Bit 0 :	Last message status	1 = OK
Bit 1 :	Last message status	1 = NOK
Bit 2 :	unused	always = 0
Bit 3 :	Communication error	1 = too many messages
Bit 4 :	Communication error	1 = identification incorrect
Bit 5 :	Communication error	1 = timeout before identification
Bit 6 :	Communication error	1 = timeout after identification
Bit 7 :	Initialisation	1 = wait for identification

One or more bits may be active at the same time.

5th Peripheral Status Bytes : additional status Byte:
only Bit 0 to Bit 2 are used

Bit 0 :	AOS frozen status	1 = frozen; 0= not frozen
Bit 1:	AOS status if AOS not configured	1 = AOS present but not configured 0 = Config. OK
Bit 2:	AOS Tdly1 for CPOD	0: Tdly1 inactive; 1 = Tdly1 active

6th Peripheral Status Bytes : Reserved for further purposes

these bytes will always show 00H .

Supply voltage value :

The measured value is transmitted as unsigned hexadecimal integer. The hexadecimal integer measurement value may be converted to a display value by means of the following formula:

Display value = ((Transmitted value / 5.3) + 0.8) * Volt

The real value may deviate from the display value within the following range :

Real value = (Display value ± 10 %) ± 0.5 Volt

The supply voltage measurement is only active after configuration programming (trouble code 52 no longer present) .

6.2.3 Read / Transmit Trouble Codes

6.2.3.1 Message structure

Tester block :

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Byte	Description	Content
1	Block length	02H
2	Block title:	
	Read Trouble Codes	12H
3	Checksum (high byte)	00H
4	Checksum (low byte)	14H

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ECU response :

The ECU response contains :

- 16 entries of the active trouble code memory together with their respective status

Before repeating the response blocks "Transmit trouble codes", the ECU performs a self test cycle of approx. 400 ms (max. 500 ms) duration. For this reason, "Wait Tester" status blocks are used to keep inter-block-times to their specified limits :

Several "Wait tester" status blocks requesting block-timeouts of 250 ms are sent before the new output of the response block.

Byte	Description	Content	
1	Block length	16H	
2	Block title Transmit trouble codes	A2H	
3	1st Trouble code byte	yyD	
4	2nd Trouble code byte	yyD	
5	3rd Trouble code byte	yyD	
6	4th Trouble code byte	yyD	
7	1st Trouble code status byte	Bits	B4 A4 B3 A3 B2 A2 B1 A1
8	5th Trouble code byte	yyD	
9	6th Trouble code byte	yyD	
10	7th Trouble code byte	yyD	
11	8th Trouble code byte	yyD	
12	2nd Trouble code status byte	Bits	B8 A8 B7 A7 B6 A6 B5 A5
13	9th Trouble code byte	yyD	
14	10th Trouble code byte	yyD	
15	11th Trouble code byte	yyD	
16	12th Trouble code byte	yyD	
17	3rd Trouble code status byte	Bits	B12 A12 B11 A11 B10 A10 B9 A9
18	13th Trouble code byte	yyD	
19	14th Trouble code byte	yyD	
20	15th. Trouble code byte	yyD	
21	16th Trouble code byte	yyD	
22	4th Trouble code status byte	Bits	B16 A16 B15 A15 B14 A14 B13 A13
23	Checksum (high byte)	ppH	
24	Checksum (low byte)	ppH	

6.2.3.2 Message contents

Trouble codes : The trouble codes are described in chapter 7 .

Trouble code status bytes : The trouble code status bytes are identical to those described in the chapter "Snapshot".

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6.2.4 Read / Transmit EEPROM Cell Contents

EEPROM contents may be read from the ECU using the "Read EEPROM Cell Contents" request.

Any number from 1 to 18 Bytes can be requested.

The requested start (and end) address may be selected within the range 0800H to 09FFh .

The ECU response in case of invalid addresses or cell numbers will be the "No Acknowledge" status block.

Tester block :

Byte	Description	Content
1	Block length	05H
2	Block title: Read EEPROM cell contents21H	
3	Number of cells to be read	nnH
4	Start address (H-Byte)	xxH
5	Start address (L-Byte)	xxH
6	Checksum (high byte)	ppH
7	Checksum (low byte)	ppH

ECU response :

Byte	Description	Content
1	Block length	08H
2	Block title Transmit EEPROM cell contents	ABH
3	Content EEPROM cell 1	eeH
... n+2	Content EEPROM cell n	eeH
n+3	Checksum (high byte)	ppH
n+4	Checksum (low byte)	ppH

6.3 Command Blocks

6.3.1 Status Block Response

The response to a command block will always be a status block. The result from execution of the command can be read from the status byte:

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Status byte	Description	
AAH	Acknowledge :	The command was correctly executed by the ECU .
55H	No Acknowledge :	The command could not be executed by the ECU .
F0H	Wait Tester :	The ECU requests a wait time from the tester.

The status "No Acknowledge" is transferred if:

- the tester block (request or command block) has been erroneous (e.g. incorrect checksum, invalid addresses)
- the tester block (request or command block) is not implemented in the ECU

The status "Wait tester" will only be transferred before another status or response block, if a block timeout would arise during processing of the tester block. This is in particular the case for the commands “Clear Trouble Codes” and "Control ECU" .

In addition, the wait time is requested once or several times before every repetition of the response blocks “Read Snapshot” and "Read Trouble Codes" .

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"Acknowledge" ECU status block :

Byte	Description	Content	
1	Block length	06H	
2	Block title		
	Transmit status block	F0H	
3	Status byte	AAH	
4	Address identification	10H	
5	Tester delay time	01H	(= 0.1 ms)
6	Tester wait time	19H	(= 250 ms)
7	Checksum (high byte)	01H	
8	Checksum (low byte)	CAH	

"No Acknowledge" ECU status block :

Byte	Description	Content	
1	Block length	06H	
2	Block title		
	Transmit status block	F0H	
3	Status byte	55H	
4	Address identification	10H	
5	Tester delay time	01H	(= 0.1 ms)
6	Tester wait time	19H	(= 250 ms)
7	Checksum (high byte)	01H	
8	Checksum (low byte)	75H	

"Wait Tester" ECU status block :

Byte	Description	Content	
1	Block length	06H	
2	Block title		
	Transmit status block	F0H	
3	Status byte	F0H	
4	Address identification	10H	
5	Tester delay time	01H	(= 0.1 ms)
6	Tester wait time	19H	(= 250 ms)
7	Checksum (high byte)	02H	
8	Checksum (low byte)	10H	

6.3.2 Transmit Status Block

Tester block :

Byte	Description	Content
1	Block length	02H
2	Block title:	
	Read status block	B0H
3	Checksum (high byte)	00H

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4	Checksum (low byte)	B2H

ECU response :

Byte	Description	Content	
	Status block:		
3	Status byte	AAH	(see 6.3.1)

6.3.3 Clear Trouble Codes

Tester block :

Byte	Description	Content
1	Block length	02H
2	Block title:	
	Erase error store	B1H
3	Checksum (high byte)	00H
4	Checksum (low byte)	B3H

ECU response :

Byte	Description	Content	
	Status block:		
3	Status byte	AAH	(see 6.3.1)

Before this status block, a wait time is always requested with status byte F0H before the trouble codes registered in the ECU are cleared. Sending of the status block begins immediately after the clearing process has ended, at the latest after t1'.

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Clearing the trouble codes will be not possible if one of the following conditions exist :

- A trouble code with status "NC" is stored in the ECU .
- A trouble code with status "NB" is stored in the ECU and one of the following conditions applies :
  - The maximum number of trouble code memory blocks has already been consumed.
  - Trouble code 52 is stored in the ECU
- Trouble code 52 is present and there has been no system configuration programming (Control ECU) within the same operating cycle

In these cases, the status byte 55H is transmitted (see 6.3.1).

Trouble codes which are detected during self diagnosis are immediately registered in the trouble code memory again. The same applies if the self diagnosis had to be stopped due to particular faults and the conditions for setting a trouble code had been present at that time.

6.3.4 End of Diagnosis

Tester block :

Byte	Description	Content
1	Block length	02H
2	Block title:	
	End diagnosis	B2H
3	Checksum (high byte)	00H
4	Checksum (low byte)	B4H

ECU response :

Byte	Description	Content
	Status block:	
3	Status byte	AAH (see 6.3.1)

The command “End diagnosis” is terminated with an "Acknowledge" status block which will be sent only once. The ECU will terminate the data transfer immediately afterwards.

6.3.5 Control ECU

6.3.5.1 Configuration Programming

The command "Control ECU” allows the system configuration of the ECU to be programmed.

Tester block :

The tester checks for programmability with the following Control ECU command :

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Byte	Description	Content
1	Block length	06H
2	Block title:	
	Control ECU	BCH
3	Control byte 1	80H
4	Execute byte 1	00H
5	Control byte 2	00H
6	Execute byte 2	00H
7	Checksum (high byte)	01H
8	Checksum (low byte)	42H

The tester performs programming with the following Control ECU command :

Byte	Description	Content
1	Block length	07H
2	Block title:	
	Control ECU	BCH
3	Control byte 1	80H
4	Execute byte 1	80H
5	Control byte 2	00H
6	Execute byte 2	00H
7	Configuration Parameter	xxH
8	Checksum (high byte)	ppH
9	Checksum (low byte)	ppH

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There are two programming modes that can be selected by the configuration parameter :

- Check Mode : correction is possible as long as the selected system configuration has not been positively verified by the ECU
  This mode is primarily intended for use with field service programmers (e.g. TECH 1).
- Forced Mode : irreversible programming performed
  This mode is intended for production line equipment only.

For all part numbers the following values for the Configuration Parameter may be selected (Note : a complete decoding table is appended in chapter 11.3) :

- independent of Bit 3 :

  Bit 7 : 0 = Check mode
  1 = Forced mode

  Bit 6 : 0 = no CUD & CUP
  1 = CUD & CUP configured

- Bit 3 = 0 :
  (system with incomplete FAD/FAP system and without passenger side suppression device)

  Bit 5 : 0 = no SAD & SAP
  1 = SAD configured
  SAP configured depending on Bit 0 :
  Bit 0 = 0 : SAP configured
  Bit 0 = 1 : no SAP (1-seat vehicles)

  Bit 4 : 0 = no BPD & BPP
  1 = BPD configured
  BPP configured depending on Bit 0 :
  Bit 0 = 0 : BPP configured
  Bit 0 = 1 : no BPP (1-seat vehicles)

  Bits 2 and 1 : see table below

Config. Parameter	FAD	FAP	AOS 1	AOS Status Lamp
x0000000B	reset to delivery mode (enter TC52)			
x0000001B	reset to delivery mode (enter TC52) or „No Acknowledge“			
xyyy000xB yyy ≠ 000B	--	--	--	*
xxxx001xB	X	--	--	*
xxxx010xB	--	X	--	*
xxxx011xB	reset to delivery mode (enter TC52) or „No Acknowledge			

X : system component configured
-- : system component not configured
* : system component not tested

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Bit 3 = 1 :
(system has both FAD and FAP and may have passenger side suppression device)

Bit 5 : 0 = no SAD & SAP
1 = SAD & SAP configured

Bit 4 : 0 = no BPD & BPP
1 = BPD & BPP configured

Bits 2 to 0 : see table below

Config. Parameter	FAP	FAD	AOS 1	AOS Status Lamp	Available only on SDM versions
xxxx1000B	X	X	--	*	
xxxx1001B	X	X	X	X	with AOS1 interface
xxxx1010B	X	X	--	X	with AOS1 interface

ECU response :

Byte	Description	Content	
	Status block:		
3	Status byte	AAH	(see 6.3.1)

The ECU responds with the status byte 55H ("No Acknowledge") in the following cases:

- The tester block "Control ECU" does not conform to the details named above.
- The trouble code 52 is no longer stored in the ECU and the parameter is not a "reset to delivery mode" value.
- A trouble code with "NB" or "NC" status is stored in the active trouble code memory block of the ECU (see chapter 7)

Before the ECU responds to a programming request with the status byte AAH, a wait time is requested with status byte F0H and the transmitted system configuration parameter and security code (if necessary) is transferred to non-volatile memory.

The last programmed state can be queried by requesting the snapshot.

The configuration programming can be repeated if required by returning into the delivery mode as long as no trouble code with "NB" or "NC" status is stored in the active trouble code memory block of the ECU.

The handling of trouble code 52 by the ECU is described in chapter 7 .

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Procedure flow :

1. Apply supply voltage to the ECU
2. Wait for at least 5 s (5 bd excitation) or 1.5 s (200 bd excitation)
   (vehicle bus stabilisation + T_{Idle})
3. Establish communication with the ECU
4. Use Control ECU command to check for programmability
5. Use Control ECU command to perform the system configuration programming
6. Clear trouble codes
7. Alternative 1 :
   Wait for at least 7 s (self diagnosis and system configuration check)
   and until at least 12 s have expired since step 1 .
   Alternative 2 :
   End communication with the ECU immediately after clearing the trouble codes.
   Wait for at least 5 s (5 bd excitation) or 1.5 s (200 bd excitation) .
   Re-establish communication with the ECU .
   Wait for at least 12 s after the communication has been ended .
   Check Mode :
   The ECU accepts the selected configuration as long as no trouble code qualifies for storage within a minimum of 5 s and a maximum equivalent to the above mentioned wait times.
   Forced Mode :
   The ECU accepts the selected configuration.
8. Evaluate snapshot :
   - Trouble Codes : no trouble codes must be entered
   - System Configuration Byte : compare with desired configuration
   - Resistance Values : check for plausibility
   - Activation Status Byte : must be reset to 00H
   - 1st Peripheral Status Byte : check for no over- or undervoltage condition present
   - 2nd Peripheral Status Byte : check for expected status
   - 5th Peripheral Status Byte : check for OK status 00H or 04H

   Systems equipped with side airbags only :
   - 3rd Peripheral Status Byte : check for OK status 01H
   - 4th Peripheral Status Byte : check for OK status 01H

N.B. : The system configuration check requires a correct and stable supply voltage to the ECU (10.5 V to 16.0 V). It will not be completely performed in power loss (e.g. ignition OFF) as well as over- or undervoltage conditions, which can cause the configuration programming to be accepted in the presence of system faults. Furthermore there might be system faults remaining undetected in these cases.

N.B. : Before retrying a failed attempt in Check Mode communication has to be terminated (equivalent to supply voltage OFF/ON).

To reconfigure after the ECU has once accepted the configuration programming it is necessary to send a Control ECU command requesting the setting of TC52 and terminate communication.

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6.3.5.2 DCO Functionality Test

Implemented on :

The command "Control ECU" allows the test of the message type 2 of the DCO .

Tester block :

The tester checks for testability with the following Control ECU command :

Byte	Description	Content
1	Block length	06H
2	Block title:	
	Control ECU	BCH
3	Control byte 1	20H
4	Execute byte 1	00H
5	Control byte 2	00H
6	Execute byte 2	00H
7	Checksum (high byte)	00H
8	Checksum (low byte)	E2H

The tester performs testing with the following Control ECU command :

Byte	Description	Content
1	Block length	06H
2	Block title:	
	Control ECU	BCH
3	Control byte 1	20H
4	Execute byte 1	20H
5	Control byte 2	00H
6	Execute byte 2	00H
7	Checksum (high byte)	01H
8	Checksum (low byte)	02H

After this command has been received, the ECU will start to output the message type 2 "Deployment Commanded" with the specified minimum number of pulses.

ECU response :

Byte	Description	Content
	Status block:	
3	Status byte	AAH (see 6.3.1)

The ECU responds with the status byte 55H ("No Acknowledge") if the tester block "Control ECU" does not conform to the details named above.

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7 Trouble Codes

The following trouble codes are generated by the Airbag ECU and transmitted in decimal form (value range 0..255) :

Code	Description		Remarks	
0	No entry in trouble code byte	L8		
4	Fault in FAD : Short to ground or battery			
5	Fault in FAP : Short to ground or battery			
6	Fault in BPD : Short to ground or battery			
7	Fault in BPP : Short to ground or battery			
8	Fault in SAD : Short to ground or battery			
9	Fault in SAP : Short to ground or battery			
2	Fault in CUD : Short to ground or battery			
3	Fault in CUP : Short to ground or battery			
31	Fault in FAD : Resistance too high			
32	Fault in FAD : Resistance too low			
33	Fault in FAP : Resistance too high			
34	Fault in FAP : Resistance too low			
35	Fault in BPD : Resistance too high		LAO	
36	Fault in BPD : Resistance too low		LAO	
37	Fault in BPP : Resistance too high		LAO	
38	Fault in BPP : Resistance too low		LAO	
77	Fault in SAD : Resistance too high		LAO	
78	Fault in SAD : Resistance too low		LAO	
87	Fault in SAP : Resistance too high		LAO	
88	Fault in SAP : Resistance too low		LAO	
101	Fault in CUD : Resistance too high			
102	Fault in CUD : Resistance too low			
103	Fault in CUP : Resistance too high			
104	Fault in CUP : Resistance too low			
43	SRS warning lamp : Short to battery		NWL	
44	SRS warning lamp : Short to ground or interrupted		NWL	
45	AOS status lamp : Short to battery		LAO	
46	AOS status lamp : Short to ground or interrupted		LAO	
52	ECU in delivery state (unconfigured)			
53	ECU configuration attempt failed			
55	ECU internal fault	L8	**NC**	
56	ECU reusability expired	L8	NC	
65	FAP activated	L8	NC	
66	FAD activated	L8	NC	DR
67	BPD (and BPP) activated	L8	NB	DR
68	SAD and CUD activated	L8	NB	DR
69	SAP and CUP activated[1]	L8	NB	DR

[1] SAP activation depends on AOS status

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Code	Description	Remarks
71	Fault SISD : Communication invalid	
72	Fault SISD : Short to battery or interrupted or short of signal line to ground	
79	Fault SISD : NOK received	
81	Fault SISP : Communication invalid	
82	Fault SISP : Short to battery or interrupted or short of signal line to ground	
89	Fault SISP : NOK received	
98	Fault in DCO : Short to ground	
90	Fault AOS1 : Communication invalid	LAO
92	Fault AOS1 : Short of signal line to battery or ground or signal line interrupted	LAO
93	Fault AOS1 : PPD fault	
95	Fault AOS1 : CPOD fault or plausibility check fault	
99	Fault AOS1 : Supply voltage too low	LPO

Abbreviations :

	driver-side	passenger-side
Front airbag	FAD	FAP
Seat Belt pretensioner	BPD	BPP
Side airbag	SAD	SAP
Curtain airbag	CUD	CUP
Side Impact Satellite	SISD	SISP

L8 : Trouble code qualified for entry into the last eight positions of trouble code memory (Positions 9 to 16 , see 6.3.2)

NB : Trouble code will lead to the "new block" mode on "Clear trouble codes" and requires an acknowledged „Clear trouble codes“ command before any further configuration programming is possible
N.B. : The ECU will not acknowledge a „Clear trouble codes“ command when a trouble code with „NB“ status is stored in the same block with trouble code 52

NC : Trouble code is not clearable via the command "Clear trouble codes" and does not allow any further configuration programming

NWL : Trouble code will not lead to an activation of the warning lamp

DR : Trouble code will lead to a deployment record (see chapter 8)

LAO : Trouble code will only lead to an activation of the warning lamp when the error is or has been present in the current operating cycle

LPO : Trouble code will only lead to an activation of the warning lamp when the error is currently present

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No trouble codes for "Supply voltage too low" or "Supply voltage too high" have been implemented. These conditions will be reported in an appropriate status bit.

The ECU stores and transmits trouble codes in the order in which they were qualified for entry into trouble code memory.

The trouble code memory is a non-volatile EEPROM within the ECU .

The ECU will only reset trouble codes when the command "Clear trouble codes" is issued. After the registration of a trouble code with "NC" status, trouble codes can no longer be reset.

In case a trouble code with "NB" status is stored when the command "Clear trouble codes" is issued, the ECU will not reset the trouble code memory but use a new block of memory area for further trouble code storage and display. Up to 7 blocks are available for trouble code storage.

Only trouble codes with "L8" status can be stored in the last eight trouble code bytes of any trouble code memory block.

Depending on the programmed system configuration (see 6.3.5.1), the following trouble codes will either not be generated, or generated with different parameters:

- The "Resistance too high" trouble codes will only be generated for configured deployment loops.
- The "Resistance too low" trouble codes will be generated with elevated threshold values for the non-configured deployment loops.
- The SISD and SISP related trouble codes will only be generated if SAD, SAP, CUD or CUP have been configured.
- The AOS related trouble code "Short to battery or interrupted or short to ground" will be generated if a non-configured AOS is detected to be present.
- The AOS status lamp trouble codes „Short to ground or interrupted" and "Short to battery" will only be generated with the AOS status lamp configured and AOS status not frozen. If AOS configured and frozen status active only the „Short to ground or interrupted" trouble code may be generated

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The trouble code 52 "ECU in delivery state (unconfigured)" is only present for an ECU in the delivery state. It does not show an existing fault, but rather serves to keep the warning lamp activated until programming of the system configuration has taken place and trouble codes have been erased for the first time after initial installation of the device in the vehicle.

As long as trouble code 52 has not been erased, the self diagnosis of the ECU peripherals as well as the deployment function of the ECU will be disabled.

When the trouble code 52 has been erased as part of the configuration programming, the unit becomes fully functional.

The trouble code 52 can only be erased, if the system configuration was programmed using “Control ECU” during the same operating cycle.

If the programming is carried out with a check mode parameter (see 6.3.5.1), the following applies :

- If the trouble code 52 is erased, the device registers an internal token, which will be stored in a non-volatile memory cell.
  Should trouble codes be registered within a certain time frame (see 6.3.5.1) after the "Clear Trouble Codes" command, the trouble codes 52 and 53 will be registered first.
  The self diagnosis of the deployment circuits will be halted again, and the ECU will in this case also not be ready for any deployment. If a "Clear trouble codes" command is issued in such an operating cycle, trouble codes will be re-entered into memory by the ECU.
  In a subsequent operating cycle, and if necessary after correcting the source of the error, the programming process must be repeated using "Control ECU".
- Should no trouble codes be registered within the time frame, the above mentioned token will be erased from the non-volatile memory. The device will thereby be made fully operational.

If the programming is carried out with a forced mode parameter, the above mentioned token will not be set.

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8 Deployment Records

To facilitate the reconstruction of possible system malfunctions a number of informations is stored with every restraint system deployment initiated by the ECU .

All trouble codes causing a deployment record will never be deleted from trouble code memory. Some of them may cause a new block of trouble code memory to be used for further fault handling should the command "Clear Touble Codes" be applied. Thus the contents of the deployment records may always be related to some entry in the trouble code memory.

Deployment records as well as hidden blocks of the trouble code memory may be read from the ECU using the "Read EEPROM Cell Contents" request.

The completeness of deployment records is dependent on the availability of external supply voltage for a sufficient time after the deployment event (approx. 1.5 s required) .

8.1 Frontal Airbag Deployment Record

This record is initiated the first time trouble code 66 has been entered into trouble code memory.

Contents :

- 1st and 2nd Trouble Code Status Byte
- Last measured FAD and FAP deployment loop resistance values
- Last measured supply voltage value
- All Peripheral Status Bytes
- Crash Algorithm Data (selected values)
- Error timer value

8.2 Other Restraint Systems Deployment Records

The ECU provides seven EEPROM memory blocks for deployment records of seat belt pretensioner or side airbag deployments. Deployment records are initiated every time a trouble code 67, 68, or 69 is entered into trouble code memory.

8.2.1 Seat Belt Pretensioner Deployment Record

This record is initiated every time trouble code 67 has been entered into trouble code memory.

Contents :

- 1st and 2nd Trouble Code Status Byte

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- Last measured BPD and BPP deployment loop resistance values
- Last measured supply voltage value
- All Peripheral Status Bytes
- Crash Algorithm Data
  (frontal / rear impact, time elapsed between first impact and deployment decision)
- Error timer value

8.2.2 Side Airbag Deployment Record

This record is initiated every time trouble code 68 or 69 has been entered into trouble code memory.

Contents :

- 1st and 2nd Trouble Code Status Byte
- Last measured SAD, SAP, CUD and CUP deployment loop resistance values
- Last measured supply voltage value
- All Peripheral Status Bytes
- Error timer value

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9 Communication Fault Recovery

9.1 Erroneous Request / Command Blocks

The ECU reacts to the reception of an erroneous tester block with the issuance of a "No Acknowledge" status block (status byte 55H). (see. 6.3.1)

9.2 Communication Termination

The ECU will terminate communication:

- on the reception of the tester command block "End of diagnosis "
- after 5 consecutive occurrences of a Byte Timeout (t_{Byte})
- if the data line is detected to be at Low-Level during the time interval t2

When the communication is terminated, the ECU performs complete restart using the internal watchdog reset function. Reestablishment of communication is only possible after observance an idle line duration of at least T_{Idle} .

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10 Quick Overview

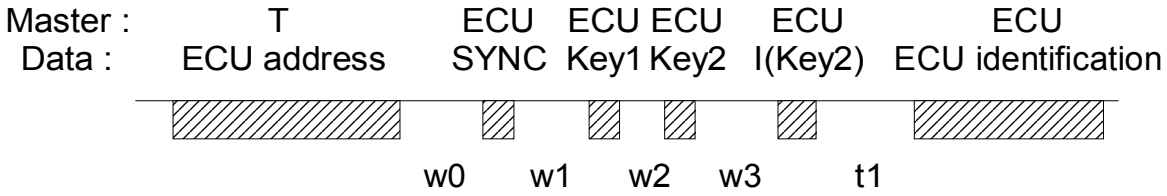
Control device address:	62H (5 Baud / 200 Baud , 8 Data bits, no parity)
Baud rate	9600 bd ± 200 bd (after SYNC byte)

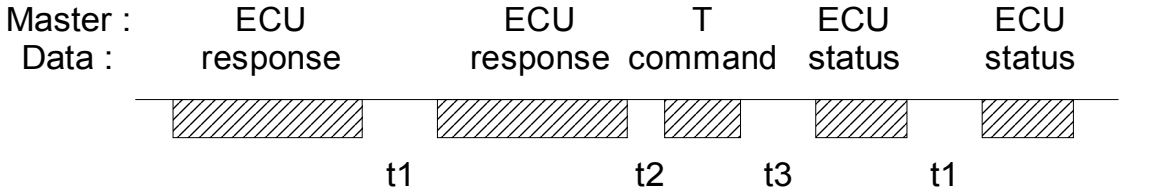
Word format:

-	Transfer:	asynchronous
-	Key bytes:	7 Data bits, parity odd, 1 stop bit
-	Data transfer	8 Data bits, no parity, 1 stop bit

Key byte 1: 52H
Key byte 2: 80H

Protocol-specific times:





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Time	Value / ms	Description
w0	>= 25 <= 50	ECU wait time between ECU address and SYNC byte (200 bd address detected)
w0	> 240 < 360	ECU wait time between ECU address and SYNC byte (5 bd address detected)
w1	>= 55 <= 195	ECU wait time between SYNC byte and key byte 1
w2	>= 1 <= 19	ECU wait time between key byte 1 and key byte 2
w3	< 200	Tester wait time between key byte 2 and its complement
t1	>= 25 <= 45	ECU interblock time : wait time before repetition of response block
t1'	>= 25 < 250	ECU interblock time : wait time before repetition of response block (after prior announcement by wait tester block)
t2	>= 1 <= 8	Tester intervention window
t3	>= 25 <= 45	ECU delay of the response block (pause between tester and ECU block)
t_{Byte}	10	Time out between bytes within a block
t_{Block}	50	Time out between two blocks
T_{Delay}	0.1	Minimum delay between two bytes (Tester Delay Time)
T_{Wait}	200	Minimum wait time for the tester (Tester Wait Time)
T_{Idle}	300	200 bd address used : Minimum Idle Line time for new stimulus after communication termination (Tester Idle Time)
$T_{Idle,PU}$	1500	200 bd address used : Minimum Idle Line time for new stimulus after ignition ON (Tester Idle Time after power-up)
T_{Idle}	2000	5 bd address used : Minimum Idle Line time for new stimulus after communication termination or ignition ON (Tester Idle Time)

Note : Several of the values given above have a tighter tolerance than prescribed by QT 12 6543 Part 1 or GME 12 543 Part 4 (ECU provided features only)

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11 Inclosure

11.1 Glossary

AOS	Automatic Occupancy Sensor
B	Binary number (numerical format)
BPD	Seat Belt Pretensioner - Driver Side
BPP	Seat Belt Pretensioner - Passenger Side
CPOD	Child Seat Presence and Orientation Detector
CUD	Curtain Airbag - Driver Side
CUP	Curtain Airbag - Passenger Side
D	Decimal number (numerical format)
DCO	"Deployment Commanded" Output
ECU	Electronic Control Unit
EEPROM	Electrically Erasable Programmable Read-Only Memory (non-volatile)
FAD	Front Airbag - Driver Side
FAP	Front Airbag - Passenger Side
H	Hexadecimal number (numerical format)
PPD	Passenger Presence Detector
SABx	Siemens Airbag system with expandable architecture
SAD	Side Airbag - Driver Side
SAP	Side Airbag - Passenger Side
SDM	Sensing and Diagnostic Module
SIS	Side Impact Satellite
T	Tester

11.2 Related Documents

65 93 14.40.42	SAB6 Opel	Satellite Sensor Communication Interface
65 93 44.40.56	SAB8x Opel	SDM Functional Specification

OPEL QT 126543 part1 05/93 issue 07. September 1993
Eigendiagnose für elektronische Systeme mit Mikroprozessoren

GME 12 543 part 4 09/94 issue 30.09.1994
Self Diagnosis of Microprocessor Controlled Electronic Systems
Communication Structure to Keyword 82

GME 12 543 part 5 05/95 issue 1
Self Diagnosis for Electronic Systems with Microprocessors
Part 5 : Physical Layer and Data Link Layer (200 Baud Initialization)

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11.3 Configuration Parameter Decoding Table (forced mode if Bit7 = 1)

Config. Parameter Binary / Hex	FAD	FAP	BPD	BPP	SAD	SAP	CUD	CUP	AOS 1	AOS Status Lamp
00000000B / 00H	**reset to delivery mode (enter TC52)**									
00000001B / 01H	reset to delivery mode (enter TC52) or „No Acknowledge“									
00000010B / 02H	**X**	**--**	**--**	**--**	**--**	**--**	**--**	**--**	**--**	*****
00000011B / 03H	X	--	--	--	--	--	--	--	--	*
00000100B / 04H	--	X	--	--	--	--	--	--	--	*
00000101B / 05H	--	X	--	--	--	--	--	--	--	*
00000110B / 06H	reset to delivery mode (enter TC52) or „No Acknowledge“									
00000111B / 07H	reset to delivery mode (enter TC52) or „No Acknowledge“									
00001000B / 08H	**X**	**X**	**--**	**--**	**--**	**--**	**--**	**--**	**--**	*****
00001001B / 09H	**X**	**X**	**--**	**--**	**--**	**--**	**--**	**--**	**X**	**X**
00010000B / 10H	--	--	X	X	--	--	--	--	--	*
00010001B / 11H	--	--	X	--	--	--	--	--	--	*
00010010B / 12H	**X**	**--**	**X**	**X**	**--**	**--**	**--**	**--**	**--**	*****
00010011B / 13H	X	--	X	--	--	--	--	--	--	*

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Config. Parameter Binary / Hex	FAD	FAP	BPD	BPP	SAD	SAP	CUD	CUP	AOS 1	AOS Status Lamp
00010100B / 14H	--	X	X	X	--	--	--	--	--	*
00010101B / 15H	--	X	X	--	--	--	--	--	--	*
00010110B / 16H	reset to delivery mode (enter TC52) or „No Acknowledge“									
00010111B / 17H	reset to delivery mode (enter TC52) or „No Acknowledge“									
00011000B / 18H	**X**	**X**	**X**	**X**	**--**	**--**	**--**	**--**	**--**	*
00011001B / 19H	**X**	**X**	**X**	**X**	**--**	**--**	**--**	**--**	**X**	**X**

Config. Parameter Binary / Hex	FAD	FAP	BPD	BPP	SAD	SAP	CUD	CUP	AOS 1	AOS Status Lamp
00100000B / 20H	--	--	--	--	X	X	--	--	--	*
00100001B / 21H	--	--	--	--	X	--	--	--	--	*
00100010B / 22H	X	--	--	--	X	X	--	--	--	*
00100011B / 23H	X	--	--	--	X	--	--	--	--	*
00100100B / 24H	--	X	--	--	X	X	--	--	--	*
00100101B / 25H	--	X	--	--	X	--	--	--	--	*
00100110B / 26H	reset to delivery mode (enter TC52) or „No Acknowledge“									
00100111B / 27H	reset to delivery mode (enter TC52) or „No Acknowledge“									

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Config. Parameter Binary / Hex	FAD	FAP	BPD	BPP	SAD	SAP	CUD	CUP	AOS 1	AOS Status Lamp
00101000B / 28H	X	X	--	--	X	X	--	--	--	*
00101001B / 29H	X	X	--	--	X	X	--	--	X	X
00110000B / 30H	--	--	X	X	X	X	--	--	--	*
00110001B / 31H	--	--	X	--	X	--	--	--	--	*
00110010B / 32H	**X**	**--**	**X**	**X**	**X**	**X**	**--**	**--**	**--**	*
00110011B / 33H	X	--	X	--	X	--	--	--	--	*
00110100B / 34H	--	X	X	X	X	X	--	--	--	*
00110101B / 35H	--	X	X	--	X	--	--	--	--	*
00110110B / 36H	reset to delivery mode (enter TC52) or „No Acknowledge“									
00110111B / 37H	reset to delivery mode (enter TC52) or „No Acknowledge“									
00111000B / 38H	**X**	**X**	**X**	**X**	**X**	**X**	**--**	**--**	**--**	*
00111001B / 39H	**X**	**X**	**X**	**X**	**X**	**X**	**--**	**--**	**X**	**X**

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Config. Parameter Binary / Hex	FAD	FAP	BPD	BPP	SAD	SAP	CUD	CUP	AOS 1	AOS Status Lamp
01000000B / 40H	--	--	--	--	--	--	X	X	--	*
01000001B / 41H	--	--	--	--	--	--	X	X	--	*
01000010B / 42H	X	--	--	--	--	--	X	X	--	*
01000011B / 43H	X	--	--	--	--	--	X	X	--	*
01000100B / 44H	--	X	--	--	--	--	X	X	--	*
01000101B / 45H	--	X	--	--	--	--	X	X	--	*
01000110B / 46H	reset to delivery mode (enter TC52) or „No Acknowledge“									
01000111B / 47H	reset to delivery mode (enter TC52) or „No Acknowledge“									
01001000B / 48H	X	X	--	--	--	--	X	X	--	*
01001001B / 49H	X	X	--	--	--	--	X	X	X	X
01010000B / 50H	--	--	X	X	--	--	X	X	--	*
01010001B / 51H	--	--	X	--	--	--	X	X	--	*
01010010B / 52H	X	--	X	X	--	--	X	X	--	*
01010011B / 53H	X	--	X	--	--	--	X	X	--	*
01010100B / 54H	--	X	X	X	--	--	X	X	--	*
01010101B / 55H	--	X	X	--	--	--	X	X	--	*
01010110B / 56H	reset to delivery mode (enter TC52) or „No Acknowledge“									
01010111B / 57H	reset to delivery mode (enter TC52) or „No Acknowledge“									

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Config. Parameter Binary / Hex	FAD	FAP	BPD	BPP	SAD	SAP	CUD	CUP	AOS 1	AOS Status Lamp
01011000B / 58H	X	X	X	X	--	--	X	X	--	*
01011001B / 59H	X	X	X	X	--	--	X	X	X	X
01100000B / 60H	--	--	--	--	X	X	X	X	--	*
01100001B / 61H	--	--	--	--	X	--	X	X	--	*
01100010B / 62H	X	--	--	--	X	X	X	X	--	*
01100011B / 63H	X	--	--	--	X	--	X	X	--	*
01100100B / 64H	--	X	--	--	X	X	X	X	--	*
01100101B / 65H	--	X	--	--	X	--	X	X	--	*
01100110B / 66H	reset to delivery mode (enter TC52) or „No Acknowledge“									
01100111B / 67H	reset to delivery mode (enter TC52) or „No Acknowledge“									
01101000B / 68H	X	X	--	--	X	X	X	X	--	*
01101001B / 69H	X	X	--	--	X	X	X	X	X	X

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Config. Parameter Binary / Hex	FAD	FAP	BPD	BPP	SAD	SAP	CUD	CUP	AOS 1	AOS Status Lamp
01110000B / 70H	--	--	X	X	X	X	X	X	--	*
01110001B / 71H	--	--	X	--	X	--	X	X	--	*
01110010B / 72H	**X**	**--**	**X**	**X**	**X**	**X**	**X**	**X**	**--**	*****
01110011B / 73H	X	--	X	--	X	--	X	X	--	*
01110100B / 74H	--	X	X	X	X	X	X	X	--	*
01110101B / 75H	--	X	X	--	X	--	X	X	--	*
01110110B / 76H	reset to delivery mode (enter TC52) or „No Acknowledge“									
01110111B / 77H	reset to delivery mode (enter TC52) or „No Acknowledge“									
01111000B / 78H	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**--**	*
01111001B / 79H	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**

Configuration parameter values are shown for check mode programming only.

Major configurations are shown in boldface.

Legend :

X : system component configured
-- : system component not configured
* : system component not tested

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11.4 Crash-Recorcing Content

Alltogether there can be 7 crash recordings for belt pret. and side airbag crashes. These crash recordings are written with an offset of 24 byte.
Therefore the side crash recording can start at the following addresses either with the content 44hex (= ASCII ‚D‘) for side airbag driver, or with the content 50hex (= ASCII ‚P‘) for side airbag passenger:
953hex, 96Bhex, 983hex, 99Bhex, 9B3hex, 9CBhex, 9E3hex

Side Crash Recording		
Offset	Description	„Size“
[hex]	.	Bytes
00	Saved code for driver (ACII'D' (= 44hex)) / passenger (ACII'P' (= 50hex))	1
01	1st Trouble Code Status Byte	1
02	2nd Trouble Code Status Byte	1
03	3rd Trouble Code Status Byte	1
04	4th Trouble Code Status Byte	1
05	For internal use	1
06	For internal use	1
07	Resistance value FAD	1
08	Resistance value FAP	1
09	Resistance value BPD	1
0A	Resistance value BPP	1
0B	Resistance value SAD	1
0C	Resistance value SAP	1
0D	Resistance value CUD	1
0E	Resistance value CUP	1
0F	Error Timer	2
11	Activation Status Byte	1
12	1st peripheral status byte: Lamp & Supply Voltage Status byte	1
13	2nd peripheral status byte: AOS1 status byte	1
14	3rd peripheral status byte: Side impact Sensor Status Driver Side	1
15	4th peripheral status byte: Side impact sensor status Passenger Side	1
16	5th peripheral status byte: additional status byte	1
17	Supply voltage value	1

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In case of Front Crash activation there will be one crash recording only. This crash recording will not be overwritten by following front activations.

SAE-mT: Front Crash Recording		
Adrress	Description	„Size“
[hex]	.	Bytes
8F3	1st Trouble Code Status Byte	1
8F4	2nd Trouble Code Status Byte	1
8F5	3rd Trouble Code Status Byte	1
8F6	4th Trouble Code Status Byte	1
8F7	Resistance value FAD	1
8F8	Resistance value FAP	1
8F9	Resistance value BPD	1
8FA	Resistance value BPP	1
8FB	Resistance value SAD	1
8FC	Resistance value SAP	1
8FD	Resistance value CUD	1
8FE	Resistance value CUP	1
8FF	Error Timer	2
901	Activation Status Byte	1
902	1st peripheral status byte: Lamp & Supply Voltage Status byte	1
903	2nd peripheral status byte: AOS1 status byte	1
904	3rd peripheral status byte: Side impact Sensor Status Driver Side	1
905	4th peripheral status byte: Side impact sensor status Passenger Side	1
906	5th peripheral status byte: additional status byte	1
907	Internal use	1
908	Supply voltage value	1
909 – 94B	Internal use	

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Alltogether there can be 7 crash recordings for belt pret. and side airbag crashes.
These crash recordings are written with an offset of 24 byte.
Therefore the belt pret. crash recording after front crash can start at the following addresses with the content 46hex (= ASCII ‚F‘):
953hex, 96Bhex, 983hex, 99Bhex, 9B3hex, 9CBhex, 9E3hex

SAE-mT: Belt Pret. Crash Recording after Front Crash		
Offset	Description	„Size“
[hex]	.	Bytes
00	saved code for front crash ACII'F' (= 46hex)	1
01	Internal use	1
02	1st Trouble Code Status Byte	1
03	2nd Trouble Code Status Byte	1
04	3rd Trouble Code Status Byte	1
05	4th Trouble Code Status Byte	1
06	Resistance value FAD	1
07	Resistance value FAP	1
08	Resistance value BPD	1
09	Resistance value BPP	1
0A	Resistance value SAD	1
0B	Resistance value SAP	1
0C	Resistance value CUD	1
0D	Resistance value CUP	1
0E	1st peripheral status byte: Lamp & Supply Voltage Status byte	1
0F	2nd peripheral status byte: AOS1 status byte	1
10	3rd peripheral status byte: Side impact Sensor Status Driver Side	1
11	4th peripheral status byte: Side impact sensor status Passenger Side	1
12	Supply voltage value	1
13-17	Internal use	

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Alltogether there can be 7 crash recordings for belt pret. and side airbag crashes.
These crash recordings are written with an offset of 24 byte.
Therefore the belt pret. crash recording after rear crash can start at the following addresses with the content 48hex (= ASCII ‚H‘):
953hex, 96Bhex, 983hex, 99Bhex, 9B3hex, 9CBhex, 9E3hex

SAE-mT: Belt Pret. Crash Recording after Rear Crash		
Offset	Description	„Size“
[hex]	.	Bytes
00	saved code for rear crash ACII'H' (= 48hex)	1
01	1st Trouble Code Status Byte	1
02	2nd Trouble Code Status Byte	1
03	3rd Trouble Code Status Byte	1
04	4th Trouble Code Status Byte	1
05	Resistance value FAD	1
06	Resistance value FAP	1
07	Resistance value BPD	1
08	Resistance value BPP	1
09	Resistance value SAD	1
0A	Resistance value SAP	1
0B	Resistance value CUD	1
0C	Resistance value CUP	1
0D	1st peripheral status byte: Lamp & Supply Voltage Status byte	1
0E	2nd peripheral status byte: AOS1 status byte	1
0F	3rd peripheral status byte: Side impact Sensor Status Driver Side	1
10	4th peripheral status byte: Side impact sensor status Passenger Side	1
11	Supply voltage value	1
12 – 17	Internal use	